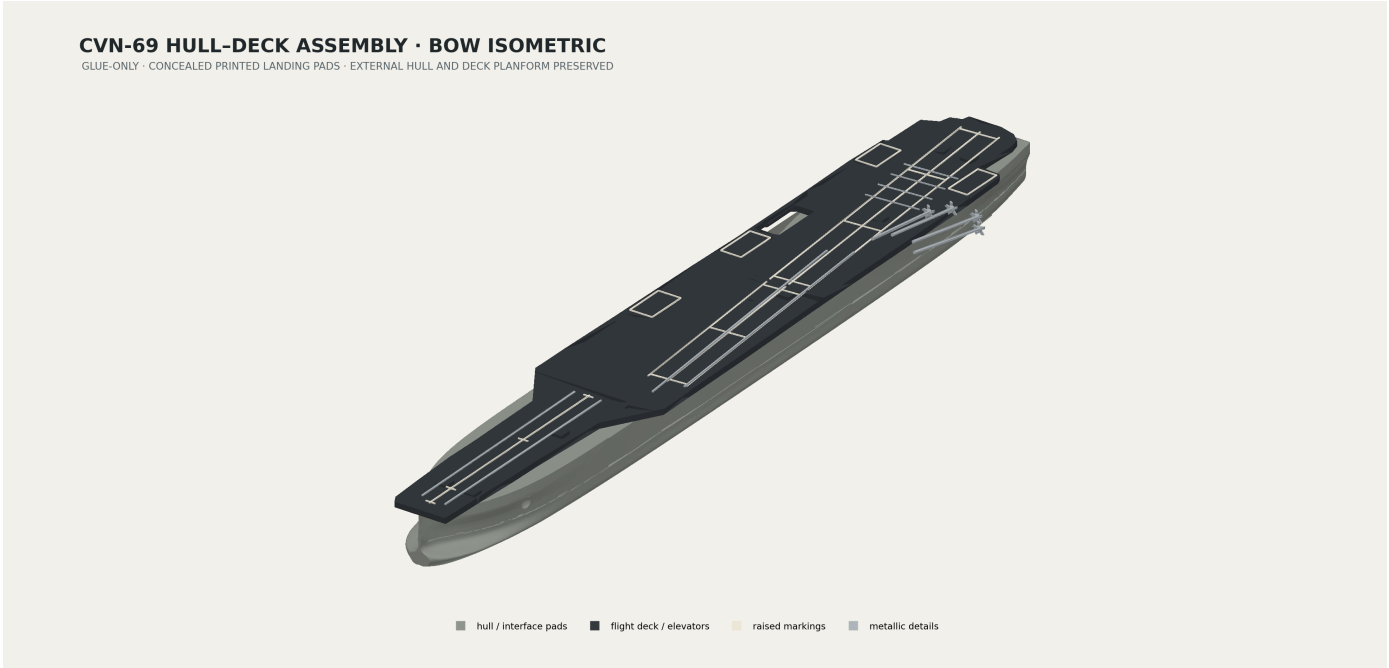


CVN-69 Milestone 2 Printing Guide

BAMBU-READY · 0.4 MM NOZZLE · GLUE-ONLY · 1:700



Print files

3MF	Contents	Validated size (mm)
Print_Plate_01_Hull.3mf	3 hull modules + shafts/struts/rudders; no props	220.06 × 177.29 × 31.43
Print_Plate_02_Deck.3mf	3 socketed deck modules	197.00 × 217.95 × 3.00
Print_Plate_03_Details.3mf	19 deck details + 12 interface pads	216.80 × 171.59 × 2.40
Print_Plate_04_Propellers.3mf	4 flat five-blade propellers	38.04 × 7.12 × 2.08
Interface_Test_Coupon.3mf	male/female clearance test	53.00 × 20.00 × 5.20

Orientation and support

- Hull modules print with their flat hull-top datum downward. Interface sockets are open to the bed; maximum socket roof span is 6.50 mm.
- Deck modules print underside down. The new sockets are open and shallow; visible deck surfaces and planform are unchanged.
- Elevators, markings, tracks, wires, and landing pads print flat. Use a brim only if your build surface needs it for the longest thin details.
- The four propellers use a common flat bed-side face and 0.60 mm blades. Print Plate 04 at 0.12 or 0.16 mm with three walls and a 3 mm brim; supports and a sprue are not required.
- Do not auto-scale. Every production STL and print 3MF is already millimetres at 1:700.

Material mapping by object name

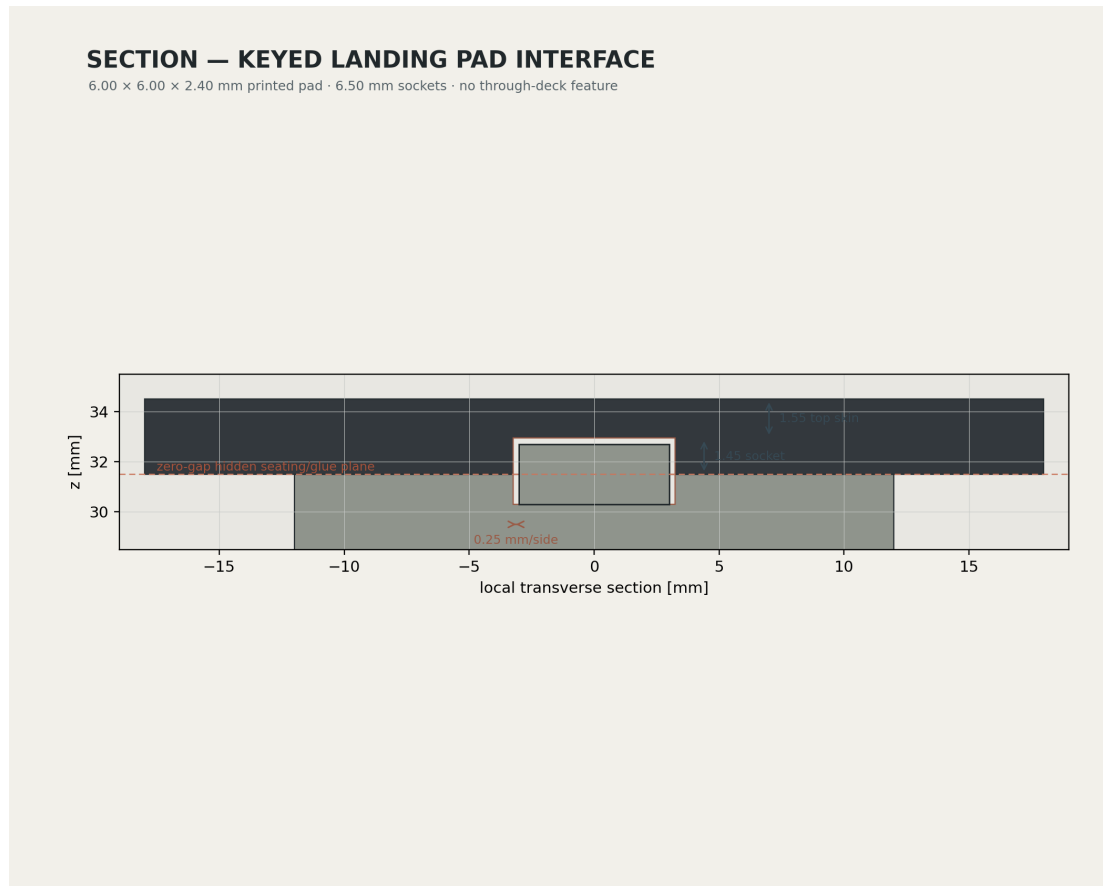
This table is also supplied as Material_Mapping.csv. AMS slot numbers are deliberately not assigned.

Object name	Material
Arresting_Wire_1	Bambu PLA Silk Silver
Arresting_Wire_2	Bambu PLA Silk Silver
Arresting_Wire_3	Bambu PLA Silk Silver
Arresting_Wire_4	Bambu PLA Silk Silver
Catapult_1_Bow_Port	Bambu PLA Silk Silver
Catapult_2_Bow_Starboard	Bambu PLA Silk Silver
Catapult_3_Waist_Inner	Bambu PLA Silk Silver
Catapult_4_Waist_Outer	Bambu PLA Silk Silver
Deck_Module_1_Bow	Bambu PLA Matte Charcoal
Deck_Module_2_Midship	Bambu PLA Matte Charcoal
Deck_Module_3_Stern	Bambu PLA Matte Charcoal
Elevator_1_Port	Bambu PLA Matte Charcoal
Elevator_2_Starboard	Bambu PLA Matte Charcoal
Elevator_3_Starboard	Bambu PLA Matte Charcoal
Elevator_4_Starboard	Bambu PLA Matte Charcoal
Hull_Module_1_Bow	Bambu PLA Matte Ash Gray
Hull_Module_2_Midship	Bambu PLA Matte Ash Gray
Hull_Module_3_Stern	Bambu PLA Matte Ash Gray
Interface_Pad_01_Port	Bambu PLA Matte Ash Gray
Interface_Pad_01_Starboard	Bambu PLA Matte Ash Gray
Interface_Pad_02_Port	Bambu PLA Matte Ash Gray
Interface_Pad_02_Starboard	Bambu PLA Matte Ash Gray
Interface_Pad_03_Port	Bambu PLA Matte Ash Gray
Interface_Pad_03_Starboard	Bambu PLA Matte Ash Gray
Interface_Pad_04_Port	Bambu PLA Matte Ash Gray
Interface_Pad_04_Starboard	Bambu PLA Matte Ash Gray
Interface_Pad_05_Port	Bambu PLA Matte Ash Gray
Interface_Pad_05_Starboard	Bambu PLA Matte Ash Gray
Interface_Pad_06_Port	Bambu PLA Matte Ash Gray
Interface_Pad_06_Starboard	Bambu PLA Matte Ash Gray
Propeller_1	Bambu PLA Silk Silver
Propeller_2	Bambu PLA Silk Silver
Propeller_3	Bambu PLA Silk Silver
Propeller_4	Bambu PLA Silk Silver
Raised_Marking_Bow_Centerline	Bambu PLA Matte Ivory White
Raised_Marking_Elevator_1_Port	Bambu PLA Matte Ivory White
Raised_Marking_Elevator_2_Starboard	Bambu PLA Matte Ivory White
Raised_Marking_Elevator_3_Starboard	Bambu PLA Matte Ivory White
Raised_Marking_Elevator_4_Starboard	Bambu PLA Matte Ivory White
Raised_Marking_Landing_Aft	Bambu PLA Matte Ivory White
Raised_Marking_Landing_Forward	Bambu PLA Matte Ivory White
Rudder_Port	Bambu PLA Matte Ash Gray
Rudder_Starboard	Bambu PLA Matte Ash Gray

Object name	Material
Shaft_1	Bambu PLA Silk Silver
Shaft_1_Strut_P	Bambu PLA Matte Ash Gray
Shaft_1_Strut_S	Bambu PLA Matte Ash Gray
Shaft_2	Bambu PLA Silk Silver
Shaft_2_Strut_P	Bambu PLA Matte Ash Gray
Shaft_2_Strut_S	Bambu PLA Matte Ash Gray
Shaft_3	Bambu PLA Silk Silver
Shaft_3_Strut_P	Bambu PLA Matte Ash Gray
Shaft_3_Strut_S	Bambu PLA Matte Ash Gray
Shaft_4	Bambu PLA Silk Silver
Shaft_4_Strut_P	Bambu PLA Matte Ash Gray
Shaft_4_Strut_S	Bambu PLA Matte Ash Gray

Interface fit and assembly readiness

Print and test the coupon before the complete assembly. The designed lateral clearance is 0.25 mm per side; the deck should seat flat before adhesive is applied.



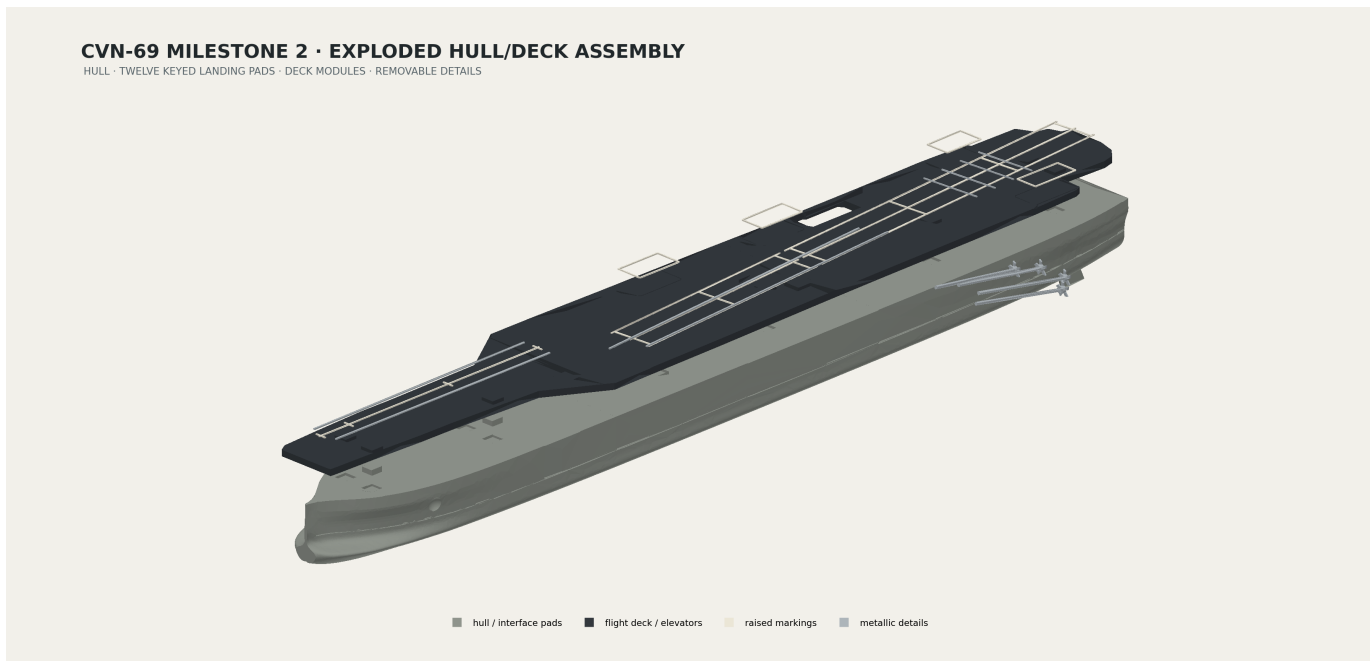
Acceptance checks

- Male feature enters without force and has no elephant-foot interference.
- Female face seats against the male base with no visible rocking or gap over 0.30 mm.
- A small amount of the intended adhesive cures without locking the parts before full seating.
- If the coupon fails, adjust slicer compensation or reprint; do not scale the production model.

Glue-only assembly sequence

Use the complete procedure in `Assembly/Glue_Only_Assembly.md`. The short sequence is:

- Assemble and cure the approved hull modules on a straight datum.
- Assemble and cure the transformed deck modules on a flat datum.
- Dry-fit all twelve pads in the hull, then lower the deck vertically and verify full seating.
- Apply a thin adhesive film only to hidden seating areas and sockets; reinstall without longitudinal sliding.
- Install elevators and raised details after the hull/deck bond has cured.



Physical qualification is complete: the 0.25 mm-per-side coupon assembled by hand and seated correctly at 100% scale with a 0.40 mm nozzle, 0.16 mm layers, three walls, 0.00 mm XY compensation, and 0.15 mm elephant-foot compensation. Production interface dimensions are frozen.